

AI Dungeon Story Generator Using GPT-2

Introduction

Artificial Intelligence has transformed the way digital content is created. Natural Language Processing (NLP) models can now understand and generate human-like text, allowing computers to assist in creative tasks such as storytelling, writing, and content creation.

The AI Dungeon Story Generator project aims to develop an AI-powered fantasy story creation platform. The system uses GPT-2, a transformer-based language model, to generate unique adventure stories according to user preferences. Users can customize their stories by selecting different characters, fantasy genres, and story prompts.

The project combines AI technology with an attractive Streamlit-based interface to provide an engaging storytelling experience. It demonstrates how modern language models can be integrated into real-world applications.

Abstract

The AI Dungeon Story Generator is an artificial intelligence-based storytelling application that creates fantasy adventure stories automatically. The project uses the GPT-2 language model from Hugging Face Transformers to generate creative and immersive narratives based on user inputs. The application provides an interactive interface where users can select a hero, choose a fantasy realm, enter a story beginning, and generate multiple story variations. Streamlit is used to build the user interface, making the application simple, interactive, and easy to use. This project demonstrates the application of Natural Language Processing (NLP) and transformer-based AI models in creative content generation.

Tools Used

- **Python** – Core development language.
- **Streamlit** – Builds the user interface.
- **Hugging Face Transformers** – Integrates the AI model.
- **GPT-2** – Generates story content.
- **CSS** – Creates the fantasy-themed design.
- **NumPy** – Library for numerical computations.

- Pandas – Library available for data handling and processing.
- Git & GitHub – Used for version control and project management.
- Visual Studio Code – Integrated Development Environment (IDE) used for coding and debugging.

Steps Involved in Building the Project

Set up the Python environment and required libraries.

Integrated GPT-2 using Hugging Face Transformers.

Created the story generation module.

Developed the Streamlit interface for user interaction.

Added prompt generation and story download features.

Tested and refined the application.

Conclusion

The AI Dungeon Story Generator successfully demonstrates the use of artificial intelligence for automated creative writing. By combining GPT-2, Hugging Face Transformers, and Streamlit, the project creates an interactive platform capable of generating unique fantasy adventures.

The project highlights the potential of NLP technologies in entertainment and creative applications. Although the current system uses GPT-2 with limitations in long-term story consistency, future improvements using advanced AI models, memory systems, and multimedia features can make the application more powerful and realistic.